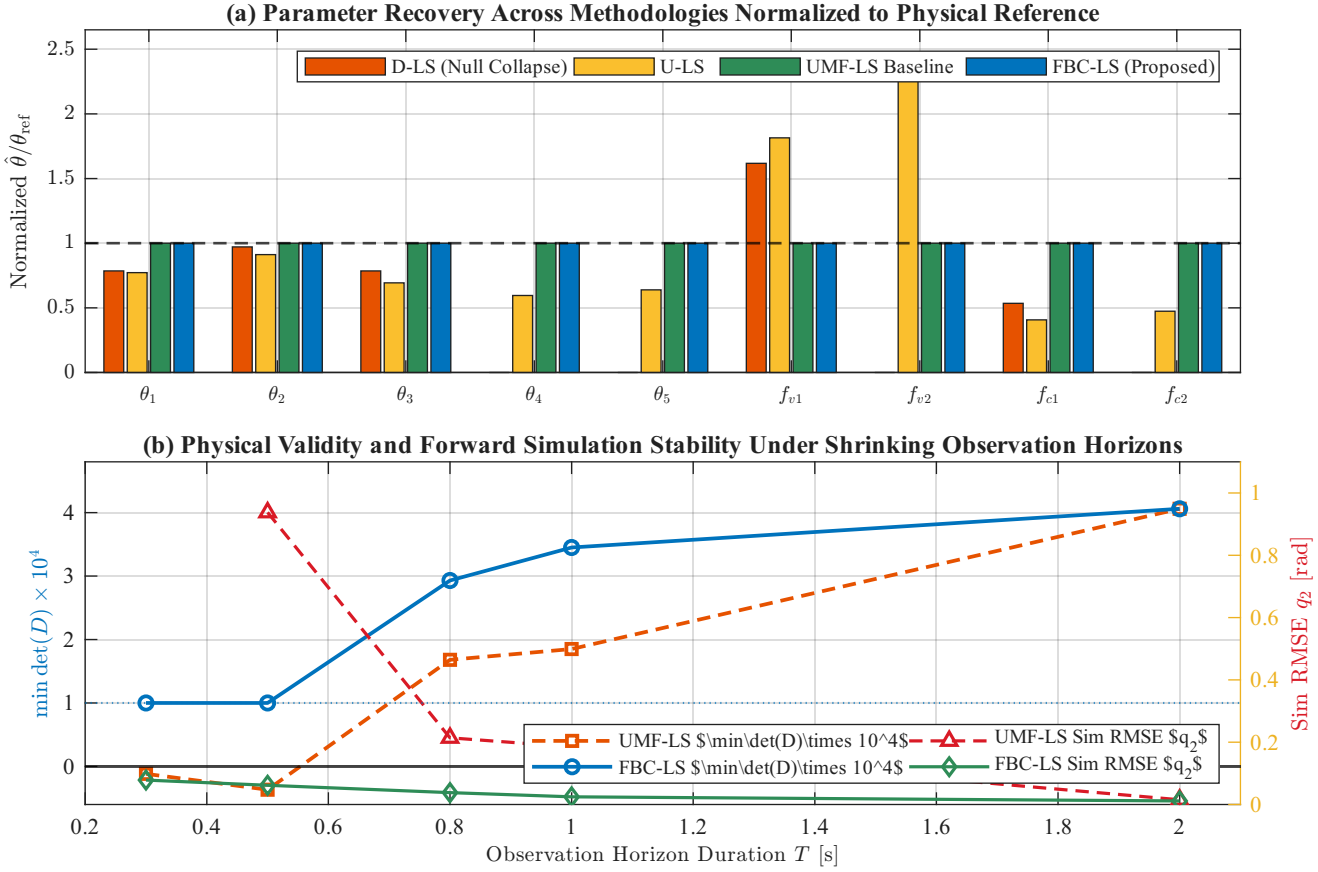
(a) Parameter Recovery Across Methodologies Normalized to Physical Reference
D-LS (Null Collapse)
U-LS
UMF-LS Baseline
FBC-LS (Proposed)
Normalized $\hat{\theta}/\theta_{\text{ref}}$
2.5
2
1.5
1
0.5
0
θ_1
θ_2
θ_3
θ_4
θ_5
f_{v1}
f_{v2}
f_{c1}
f_{c2}
(b) Physical Validity and Forward Simulation Stability Under Shrinking Observation Horizons
$\min \det(D) \times 10^4$
Sim RMSE q_2 [rad]
UMF-LS $\min\det(D)\times 10^4$
UMF-LS Sim RMSE q_2
FBC-LS $\min\det(D)\times 10^4$
FBC-LS Sim RMSE q_2
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Observation Horizon Duration T [s]